

Course Code	: 2304CS301	Date	: 11-10-2025
Course Name	: Computer Network	Duration	: 150 Minutes
		Total Marks	: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 (A) Write a short note on LAN, MAN and WAN. **4**

(B) Define the following terms: **3**

- (i) Delay
- (ii) Throughput
- (iii) Packet Loss

OR

List the applications of computer networks.

(C) List the types of network topologies and discuss ring and star topologies with suitable diagrams. **7**

OR

List the types of network topologies and discuss tree and hybrid topologies with suitable diagrams.

Q.2 (A) Discuss piconet and scatternet with a suitable figure. **4**

(B) Draw a neat and clean diagram of a coaxial cable. **3**

OR

Draw a neat and clean diagram of a fiber-optic cable.

(C) What is unguided media? Explain radio waves and microwave along with their characteristics. **7**

OR

Explain twisted-pair cable, its types and also discuss its advantages and disadvantages.

Q.3 (A) Compare circuit switching vs packet switching. **4**

(B) Write a short note on half-duplex mode. **3**

OR

Write a short note on full-duplex mode.

(C) Explain the OSI reference model with a suitable diagram. **7**

OR

Explain the TCP/IP protocol suite with a proper figure.

- Q.4 (A)** Explain a file server and a web server with a suitable figure. **4**
(B) Write a short note on a Hub. **3**

OR

Write a short note on a Gateway.

- (C)** Explain the bridge networking device along with its types. **7**

OR

Explain the switch networking device along with its types.

- Q.5 (A)** Define the following terms: **4**
(i) SMTP
(ii) FTP
(iii) DNS
(iv) HTTP

- (B)** Identify the class of the following IP addresses: **3**
i. 181.168.10.1
ii. 10.10.10.0
iii. 249.158.75.56

OR

- i. 53.210.21.101
ii. 172.30.56.241
iii. 230.133.65.60

- (C)** Discuss IPv4 header format with a suitable diagram. **7**

OR

Describe the WWW and web hosting with a proper example.
